



SIMATS Online Programming Lab

DS PROGRAMMING



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QUESTIONS

10 / 10 completed

Linear Search on Array

LINEAR SEARCH ON ARRAY

"Develop a C program to perform Linear Search on an array of N integers. Accept array size dynamically

Sample Input/Output

Enter size: 5

Array: 10 20 30 40 50

Enter target: 40

Comparison Trace:

10 ≠ 40

20 ≠ 40

30 ≠ 40

40 = 40

Target Found at Position 4

Comparisons Made = 4

Test Cases

- Element at first index
- Element at last index

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 int main() {
5     int n;
6     printf("Enter size: ");
7     if (scanf("%d", &n) != 1 || n <= 0) {
8         printf("Invalid size.\n");
9         return 0;
10    }
11
12    int *arr = (int *)malloc(n * sizeof(int));
13    if (arr == NULL) {
14        printf("Memory allocation failed.\n");
15        return 0;
16    }
17
18    printf("Array: ");
19    for (int i = 0; i < n; i++) {
20        scanf("%d", &arr[i]);
21    }
```

OUTPUT

```
Enter size: Array: Enter target: 10 ≠ 40
20 ≠ 40
30 ≠ 40
40 = 40
Target Found at Position 4
Comparisons Made = 4
```

INPUT

5

10 20 30 40 50

40

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